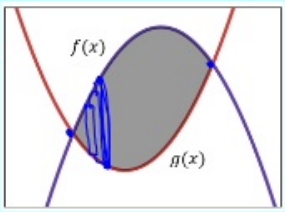
CHA-5.A.1 Areas of regions in the plane can be calculated with definite integrals
CHA-5.A.2 Areas of regions in the plane can be calculated using functions of either x or y.
CHA-5.A.3 Areas of certain regions in the plane may be calculated using a sum of two or more definite integrals or by evaluating a definite integral of the absolute value of the difference of two functions.

8-4 to 8-6 Area Between Curves

We are interested in finding the area between two curves. Consider the functions at the right:

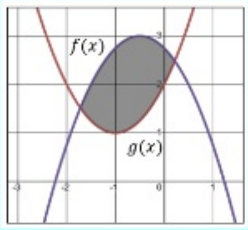


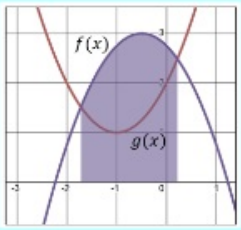
To find the area between these two curves, we can use the same approach we used for Riemann Sums.

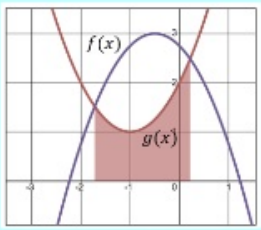
We will need two pieces of information to find the area between the two curves:

1. Where do the two curves intersect.
2. Which curve has the higher value.

With a slight modification, we can change the concept of finding the area of a region <u>under</u> a curve (that lies above the x-axis) to finding the area of a region <u>between</u> two curves.



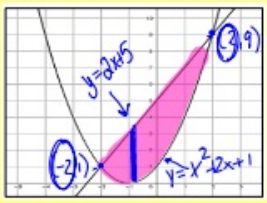




The area between the graphs of $f(x)$ and $g(x)$	=	The area under the graph of $f(x)$	−	The area under the graph of $g(x)$

Area of a Region between two curves	If f and g are continuous on $[a, b]$ and $g(x) \leq f(x)$ for all x in $[a, b]$, then the area of the region bounded by the graphs of f and g and the vertical lines $x = a$ and $x = b$ is $$A = \int_a^b \lvert f(x) - g(x) \rvert \, dx$$ Top Curve − Bottom curve

Example 1: Find the area bounded by the curves $f(x) = x^2 + 2x + 1$ and $g(x) = 2x + 5$



$$\text{Area} = \int_{-2}^{2} \left[2x+5-(x^2+2x+1)\right] dx$$

$$= \int_{-2}^{2} (2x+5-x^2-2x-1)\, dx$$

$$= \int_{-2}^{2} (-x^2+4)\, dx$$

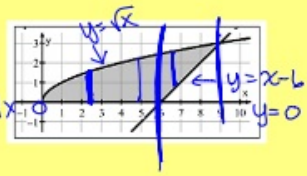
$$= -\frac{1}{3}x^3 + 4x \Big|_{-2}^{2}$$

$$= -\frac{8}{3} + 8 - \left(\frac{8}{3} - 8\right)$$

$$= -\frac{16}{3} + 16 = -\frac{16}{3} + \frac{48}{3} = \boxed{\frac{32}{3}}$$

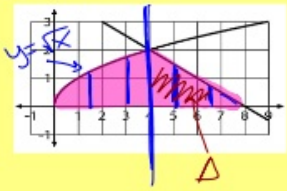
Example 2: Set up, but do not evaluate, the integral(s) that can be used to find the area in the first quadrant that is bounded by $y = \sqrt{x}$ and $y = x - 6$.

$$\text{Area} = \int_0^6 (\sqrt{x} - 0)\,dx + \int_6^9 \left[\sqrt{x} - (x-6)\right]dx$$



Example 3: The graphs of $f(x) = \sqrt{x}$ and $g(x) = -\frac{1}{2}x + 4$ are below. Determine the area enclosed by these graphs and the x-axis.

$\text{Area}\triangle = \frac{1}{2}(4)(2) = 4$



$$\text{Area} = \int_0^4 \sqrt{x}\,dx + \underbrace{\int_4^8 -\frac{1}{2}x+4}_{\text{Area}=4}$$

$$= \int_0^4 x^{1/2}\,dx + 4$$

$$= \frac{2}{3}x^{3/2}\Big|_0^4 + 4$$

$$= \frac{16}{3} - 0 + 4$$

$$= \frac{16}{3} + \frac{12}{3} = \boxed{\frac{28}{3}}$$

$4^{3/2} = (\sqrt{4})^3$

Sometimes it is better to integrate with respect to y as opposed to with respect to x. EQ must be x = $A(y) = \int_c^d [f(y) - g(y)]\,dy$ $y=c$ $y=d$

Caution: Remember the limits of integration must be y-values!

Right curve - left curve

Example 4: Find the area enclosed by the line $y = x - 1$ and the parabola $y^2 = 2x + 6$

SOLVE FOR X

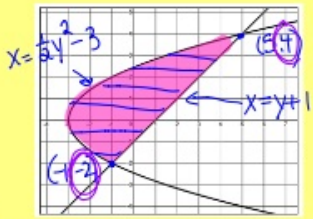
$y = x - 1$

$y + 1 = x$

$y^2 = 2x + 6$

$\frac{y^2}{2} - \frac{6}{2} = \frac{2x}{2}$

$\frac{1}{2}y^2 - 3 = x$



$$\text{Area} = \int_{-2}^{4} \left[y + 1 - \left(\frac{1}{2}y^2 - 3\right)\right] dy$$

$$= \int_{-2}^{4} \left(y + 1 - \frac{1}{2}y^2 + 3\right) dy$$

$$= \int_{-2}^{4} \left(-\frac{1}{2}y^2 + y + 4\right) dy$$

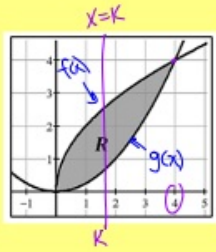
$$= \left[-\frac{1}{6}y^3 + \frac{1}{2}y^2 + 4y\right]\Big|_{-2}^{4}$$

$$= \frac{-64}{6} + 8 + 16 - \left(\frac{8}{6} + 2 - 8\right)$$

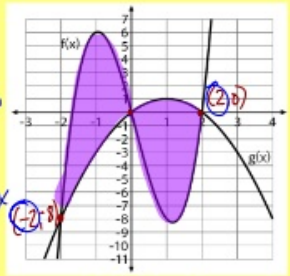
$$= \boxed{18}$$

<u>Example 5:</u> (Calculator Active) Let R be the region bounded by the graphs of $f(x) = 2\sqrt{x}$ and $g(x) = \frac{x^2}{4}$ as shown in the figure below. If the vertical line $x = k$ divides R into two regions of equal area, what is the value of k? <u>SETUP ONLY</u>

$$\int_0^k (f(x) - g(x))\,dx = \int_k^4 (f(x) - g(x))\,dx$$



Example 6: Find the area of the regions bounded by the graphs of $f(x) = 3x^3 - x^2 - 10x$ and $g(x) = -x^2 + 2x$



$$\text{Area} = \int_{-2}^{0} 3x^3 - x^2 - 10x - (-x^2 + 2x)\,dx$$

$$+ \int_{0}^{2} -x^2 + 2x - (3x^3 - x^2 - 10x)\,dx$$

$$= \int_{-2}^{0} (3x^3 - x^2 - 10x + x^2 - 2x)\,dx + \int_{0}^{2} (-x^2 + 2x - 3x^3 + x^2 + 10x)\,dx$$

$$= \int_{-2}^{0} (3x^3 - 12x)\,dx + \int_{0}^{2} (-3x^3 + 12x)\,dx$$

$$= \left(\frac{3}{4}x^4 - 6x^2\right)\Big|_{-2}^{0} + \left(-\frac{3}{4}x^4 + 6x^2\right)\Big|_{0}^{2}$$

$$= 0 - 0 - (12 - 24) + -12 + 24 - (0 + 0)$$

$$= 12 + 12 = \boxed{24}$$

Example 7: (Calculator Active) Sketch the region(s) enclosed by the following graphs and find the total area enclosed between the curves.

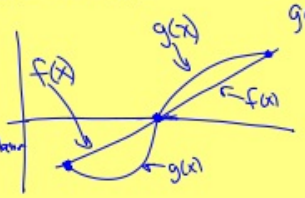
$$f(x) = e^{0.2x} - 2.6 \text{ and } g(x) = \cos(x)$$

$f(x)=Y_1$
$g(x)=Y_2$

X-values of pts. of int

$X=2.674299541\ldots = A$
$X=4.643603712\ldots = B$
$X=6.395855163\ldots = C$

Stored in Calculator



$$\text{Area} = \int_A^B (f(x)-g(x))\,dx + \int_B^C (g(x)-f(x))\,dx$$

$$= 0.80218324\ldots$$

$$\approx 0.802$$